RESEARCH BRIEFING

Integrating BBNJ standardized batch identifiers into Darwin Core

Fields, extensions, EML, INSDC/BioSamples, and linking sequences to OBIS/GBIF by batch ID

Sources: BBNJ Agreement & PrepCom/CHM studies; Humphries, Rabone et al. (Springer 2025); Frontiers CHM capacity paper (2026); DwC / DwC-DP / Latimer Core; GBIF EML profile, extensions registry, projectId work; UK guidance; INSDC notes; DwC/GGBN specs

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1 Executive summary

The BBNJ Agreement requires a **"BBNJ" standardized batch identifier** (BSBI / BBNJ Identifier): a Clearing-House Mechanism (CHM) tag for a **batch** of marine genetic resource (MGR) collection in areas beyond national jurisdiction — not a per-specimen or per-sequence ID. It must travel with samples and digital sequence information (DSI) through notifications and support **aggregate** access reporting: a traceability end-point, not full track-and-trace.

Format is still undecided by the Conference of the Parties. There is **no ratified Darwin Core term** for it. State-of-the-art guidance (Rabone et al., 2025) says capture in OBIS/GBIF via **existing Darwin Core fields "may be achievable,"** while Latimer Core is better for **collection** holdings and INSDC likely needs more explicit UI fields. That advice is directionally right on reuse — but it under-specifies **which** fields, and it does not solve **queryability**.

For biodiversity portals the product goal is concrete: **filter occurrences (and events) in OBIS/GBIF by batch identifier** — and, equally, **join those records to INSDC/EBI BioSamples and sequence accessions that carry the same ID.** DwC field choice is not independent of what GenBank/ENA/DDBJ decide: if INSDC only accepts free-text cruise fields, DwC must still be typed and indexed, and a linking layer must normalise both sides. Dataset-only EML, untyped `eventID` overload, and eMoF-as-identifier all fall short.

> **Recommendation (authoritative)**
>
> **Do not** open with a BBNJ-specific DwC core term (`bbnjBatchID`). TDWG prefers **generic** identifier concepts (cf. `recordedByID`, `projectID`) with schemes in usage guidance — Tim Robertson's framing for the public issue.

Canonical model: DwC-DP `event-identifier` / `survey-identifier` (typed rows; multiple IDs per event; `identifierType=bbnjBatch`).
DwC-A path (pick one, discuss at TDWG): (C1) **update** the GBIF Identifier extension so it is not taxon-only and documents `identifierType` for Event/Occurrence; or (C2) sandbox a **new** Event Identifier extension mirroring DwC-DP. Prefer not "use Alternative Identifiers as-is."
Why C1 is suddenly credible: GBIF's DwC-DP → DwC-A-shaped ingest already fans `event-identifier` (and other `*-identifier` tables) out to rowType `http://rs.gbif.org/terms/1.0/Identifier` (`IdentifierMapping`). Updating that extension aligns publishing docs with what ingest already does.
Processing + downloads: project the typed batch ID onto Event **and** Occurrence search docs; for downloads prefer a **dedicated identifier file** (plus optional Occurrence column for simple CSV). Do not rely on extension rows alone.
Keep GGBN Permit for Nagoya/IRCC; align BioSamples to the same type string; MIxS is not the home (unsequenced MGR still need the ID).

2 What the identifier is (and is not)

2.1 Treaty role

Under arts. 12, 14 and 16:

- Pre-collection notification → CHM **automatically generates** the batch identifier (art. 12(3)).
- The same ID is carried in post-collection and utilization notifications.
- Repositories/databases under Party jurisdiction must accompany MGR/DSI with the ID where required (art. 14(3)).
- Biennial **aggregate** access reports are linked to batch identifiers (art. 12(7)) — not specimen-level use.

Negotiators chose **batch** rather than **cruise**: one ID may map to a cruise, leg, AUV deployment, or project once COP procedures fix the grain. It is deliberately lighter than PIP-style track-and-trace and closer to MSR cruise practice than to CBD/Nagoya IRCC identifiers.

2.2 What is not decided

UK guidance states the **format has not yet been determined**. PrepCom/CHM materials leave generation procedures, machine-readability, resolution, and exact scope to COP/ABSC/STB. Biodiversity networks can design fields and indexes now, but should not freeze validators on a guessed syntax. Harmonisation with IGSN/SESAR-style batch registration is recommended in recent capacity work (Frontiers, 2026) to avoid colliding identifier ecosystems.

2.3 Relation to scientific identifiers

Concept	Typical grain	Role
BBNJ batch ID	Collection batch / notification	Legal–administrative link to CHM & aggregate reports
Cruise / EXPOCODE	Cruise / leg	Scientific practice; may coincide with or nest under a batch
`eventID`	Sampling event in one DwC dataset	Structural hierarchy inside a published dataset
`occurrenceID` / `materialSampleID`	Observation or sample	Scientific identity
INSDC accession / BioProject	Sequence / study	DSI identity; should **carry** the batch ID
EML `project` / `alternateIdentifier`	Dataset / package	Resource metadata — not occurrence search

Table 1: Grain and role of related identifiers.

Conflating BBNJ batch ID with `eventID` breaks Event-core hierarchies already used in OBIS and breaks cross-dataset uniqueness expectations. Treat it as an **additional**, **typed** identifier.

2.3.1 Why not put the batch ID in `eventID`?

This is the most common shortcut (make the cruise/batch parent's `eventID` equal the CHM string). It looks attractive with an event index — filter `eventID = batch` — but it conflates three different jobs:

1. **Structural key** inside a dataset (`eventID` / `parentEventID` tree; joins to occurrences and extensions).
2. **Scientific / operational label** (local cruise code, station code, or other human-readable campaign labels — better as **typed** alternate identifiers, not as the structural key).
3. **External legal–administrative identifier** issued by the CHM (and mirrored in INSDC/BioSamples).

`eventID` is defined only for (1): "an identifier for the set of information associated with a dwc:Event … specific to the data set" or a GUID. It is the hinge of the Event-core star schema. Overloading it with (3) fails in concrete ways:

- **Publishers already have their own `eventID`s.** Replacing a parent's `eventID` with a CHM string forces a rewrite of every child's `parentEventID`, breaks continuity with publications that cited the old codes, and collides when the same cruise already has a scientific cruise ID **and** a BBNJ ID (common: negotiators deliberately allowed batch ≠ cruise).
- **One string, many roles → untyped discovery.** Searchers cannot tell a CHM ID from a local cruise code. INSDC's "put it where cruise identifiers go" has the same flaw; DwC should not copy it.
- **Hierarchy grain.** A batch may sit above several cruises, or one cruise may split into several batches (legs, AUV recoveries). `eventID` is 1:1 with an Event row; external IDs are often many:1 or 1:many relative to scientific events — that is why DwC-DP has a separate `event-identifier` table.
- **Aggregation / cross-dataset joins.** DwC-DP #159's whole point: semantic DwC IDs collide across datasets and must not be the only relational key. Making the BBNJ string the `eventID` couples a **global** legal ID to a **dataset-scoped** structural field — the opposite of the emerging `*_pk` / `*_fk` vs `eventID` split.
- **GBIF inheritance gap.** Even if OBIS walks parents, GBIF often does not promote parent `eventID` onto occurrences. Children keep station `eventID`s; the batch string on the parent is invisible at occurrence search unless denormalised as its **own** field.
- **Wrong equality for BioSamples.** The join to INSDC is "same batch string on both sides," not "BioSample.eventID = DwC.eventID." Samples need `materialSampleID` = BioSample accession **plus** a shared batch attribute.

What to do instead with your prototype: index a typed `bbnjBatchID` (or `event-identifier` row) on the event document/`event.parquet`; filter on that; join children and occurrences with the unchanged scientific `eventID`. Same retrieval power, without burning the structural key.

2.3.2 Precedents that inspire the right shape

Precedent	Lesson for BBNJ
DwC `fieldNumber` vs `eventID`	Field-note / operational labels are **not** the Event's structural ID (GBIF Humboldt survey guide; IPT quality checklist).
DwC `projectID` / `projectTitle` (ratified 2025–26, tdwg/dwc#532)	Record-level project attribution got a **new term**, not overload of `eventID`. Closest DwC pattern; BBNJ is similar (external GUID on Event) but **not** the same semantics as funder/project IDs — do not silently reuse `projectID` for CHM batches.
DwC-DP `event-identifier` / `survey-identifier`	First-class table: `identifier` + `identifierType` (+ IRI/source) linked by `eventID` → Event. Designed for multiple typed IDs per event (cruise + DOI + BBNJ).
DwC-DP #159 / #150	Separate **structural** keys from **semantic** DwC identifiers so aggregates and joins stay faithful.
GBIF Identifier extension; Latimer Core `Identifier`	Typed alternate IDs beside core records / collections.
Plant Treaty GLIS DOI vs accession ID	Legal/global PID **alongside** local scientific accession — GLIS FAQ: DOI does not replace accession identifiers.
`recordedByID` / ORCID vs `recordedBy`	Machine ID in a dedicated field; human label stays separate (tdwg/dwc#102 lineage).
GGBN Permit extension	ABS/legal metadata as extension rows, not stuffed into specimen IDs — see dedicated section below.
ABSCH / IRCC Unique Identifier	Cautionary tale + partial reuse: certificate-level UID via `permitURI`; low uptake / weak machine-readability — do **not** copy that failure mode.
WIGOS station IDs (GOOS)	External platform registry IDs kept distinct from sampling-event IDs — same dual-ID problem OBIS flags for BBNJ + WIGOS in the README.

Table 2: Precedents: external / typed IDs kept **beside** structural IDs.

2.3.3 Ongoing work and discussions (repos / issues)

- **gbif/dwc-dp #159** (closed Jun 2026) — structural `*_pk`/`*_fk` vs semantic `eventID`; directly motivates not overloading `eventID` with global legal IDs. Related: #150 weak foreign keys.
- **gbif/dwc-dp** table schemas — `event-identifier.json`, `survey-identifier.json` (typed identifiers); open housekeeping (#139 production plan); no BBNJ-specific issue yet.
- **tdwg/dwc #532 / #527** — `projectID` / `projectTitle` (and funding terms) completed; pattern for external IDs on Event. #671 `fieldNumber` in DwC-DP review. **gbif/dwc-dp #112** — "where to put human-readable site identifiers?" — community repeatedly refuses to burn `eventID` for alternate labels.
- **gbif/pipelines #836** (closed) / **gbif/rs.gbif.org #115** (open) — promote `projectId` to **occurrence** search, not only EML; same "dataset metadata ≠ record filter" lesson for BBNJ.
- **iobis/pipeline** — `separate-tables` branch (event/occurrence/extension Parquet); little public BBNJ issue traffic yet; README Applications stubs for BBNJ SBI and WIGOS.
- **No TDWG issue titled for BBNJ batch ID** was found — green field for a term or `identifierType` vocabulary once CHM format stabilises.

Interim temptation: map BBNJ → `projectID` because the term exists. Resist unless CHM explicitly aligns batch≡project: `projectID` already means funder/research projects (Ocean Census, RCN, …). Prefer typed `event-identifier` / dedicated term / Identifier extension with `identifierType=bbnjBatch`.

2.3.4 ABSCH, Nagoya, and the GGBN Permit extension — what to reuse

2.3.4.1 ABSCH / IRCC (what it is, and why it is only a partial model)

Under the Nagoya Protocol, Parties that require PIC publish a permit (or equivalent) to the ABS Clearing-House; that publication **constitutes** an Internationally Recognized Certificate of Compliance (IRCC). Each IRCC carries an **ABSCH Unique Identifier** (UID). Minimum IRCC content (when not confidential): issuing authority, date, provider, UID, PIC grantee, subject-matter/resources, confirmation of PIC and MAT, commercial/non-commercial use. The UID attaches to the **certificate** (proof of lawful access), not to each physical sample — the same "tag the administrative act, not every organism" idea as the BBNJ batch ID.

Documented problems (reuse as anti-patterns):

- **Low uptake** — on the order of a few thousand IRCCs globally (≈5k as of May 2024 in Humphries et al.), concentrated in few Parties; most non-commercial research.
- **Weak machine-readability / propagation** — literature repeatedly flags that the ABSCH UID is hard to use as a join key into OBIS/GBIF/INSDC (Rabone et al. 2019; Rohden et al. 2020; Scholz et al. 2022; Humphries et al. 2025). Checkpoints and IRCC links rarely appear on sequence or occurrence records at scale.
- **Track-and-trace ambition vs reality** — Nagoya's checkpoint machinery aims further down the R&D chain than BBNJ's aggregate-report model; BBNJ negotiators deliberately chose a **lighter** end-point. Copying Nagoya's **legal** shape without fixing **data** propagation would repeat the failure.

What is still reusable from ABSCH/Nagoya:

- **Certificate/notification-level ID, not specimen-level ID** — same grain philosophy as BBNJ.
- **Resolvable URI to the clearing-house record** as the primary machine link (`https://absch.cbd.int/...` today; analogous CHM URL tomorrow).
- **Revision policy lesson** — ABSCH UIDs can carry revision numbers on amendment; BBNJ keeps the batch ID stable while notification metadata updates (Humphries et al.). DwC should store the **stable** ID and optionally a separate CHM record URL/version.
- **Dual regimes** — ABNJ batches may still interact with Nagoya when materials later enter national jurisdiction, or when TK obligations apply. Records may legitimately carry **both** an IRCC UID and a BBNJ batch ID — another reason not to burn `eventID` for either.

2.3.4.2 GGBN Permit extension (the concrete DwC reuse path for ABS)

The Global Genome Biodiversity Network defined a **Permit** vocabulary and registered it as a GBIF DwC-A extension (`http://data.ggbn.org/schemas/ggbn/terms/Permit`, issued 2022-08-08; also ABCD-GGBN). Purpose: attach **legal/administrative documents** to **MaterialSample** (tissue/DNA/specimen) records in a 1 :many star — collecting permit, export/import, MTA, MoU, **and IRCC** — without overloading `catalogNumber` / `materialSampleID` / `eventID`.

Extension properties (all under `ggbn:`):

Property	Req.	Role
`permitType`	yes	Controlled vocab: Collecting Permit, Export/Import, MTA, MoU, **Internationally Recognized Certificate of Compliance**, Patent, Data use, Other, …
`permitStatus`	yes	Permit available / not required / not available / Unknown (post-2014-10-12 Nagoya entry: Unknown discouraged)
`permitStatusQualifier`	no	Why not required / unknown — e.g. "pre-Nagoya", "no national requirement"
`permitURI`	no	Resolvable link to the permit or ABSCH IRCC record
`permitText`	no	Human-readable excerpt or details

Table 3: GGBN Permit extension fields (rs.gbif.org / GGBN wiki).

How it is meant to be published: MaterialSample core (GGBN preferred) or Occurrence with Material Sample extension; many Permit rows per sample. GGBN marks permit fields **highly recommended** once Nagoya is in force. Wiki note on `permitURI`: "One will eventually need a direct link to ggbn:IRCC in the future" — i.e. they already foresaw first-class IRCC linking beyond a generic URI.

What this gives BBNJ / OBIS:

1. **Proven pattern** for "legal ID beside scientific ID" in DwC-A — exactly the separation argued for batch IDs.
2. **Immediate dual-use:** for ABNJ work that also has Nagoya touchpoints, publish `permitType=IRCC` + `permitURI=https://absch.cbd.int/…` on the material/sample; publish BBNJ batch ID as a **typed event identifier** (or future permitType value / dedicated term) — do not merge them.
3. **Extension-as-carrier template** if CHM needs more than one string (notification URL, status, "permit not required for ABNJ multilateral mechanism", etc.): a thin BBNJ extension can mirror GGBN's five-field shape (`type`, `status`, `qualifier`, `uri`, `text`) rather than inventing a novel shape.
4. **Caveats for OBIS occurrence query:** GGBN Permit is **sample-centric** and **not** a first-class OBIS/GBIF occurrence search facet today. Reusing it for IRCC/MTA is right; relying on it **alone** for "filter all OBIS occurrences by BBNJ batch ID" is not — still need event-level typed ID + index promotion (your event index path).

Reuse verdict: ABSCH + GGBN

Reuse GGBN Permit for Nagoya/IRCC/MTA documents on materials. **Learn from ABSCH** (stable certificate ID + URI; avoid non-machine-readable / non-propagated UIDs). **Do not treat IRCC UID or GGBN Permit as a substitute for a typed BBNJ batch field on events** — different legal instrument, different grain (certificate/sample vs collection batch), and Permit rows are not occurrence search keys. Where both apply, carry both.

3 State of the art (what has actually been said)

3.1 Policy and CHM operationalisation

- **Agreement text** — normative generation and use.
- **PrepCom / consolidated CHM technical study (2026)** — CHM generates and maintains identifiers but is **not** mandated to track-and-trace all downstream uses; lists OBIS, GBIF, INSDC among relevant infrastructures; notes an **OBIS–GenBank pilot** for batch-ID management and linking; INSDC expects to accept the ID in **free-text cruise-like metadata** without enforcing compliance.

- **National guidance** (e.g. UK) — operational flow; format unknown; associate ID with all project MGR/DSI.

3.2 Scientific / legal literature (Springer 2025 and after)

Humphries et al. (data management & identifier) and Rabone et al. (scientists / end users) are the most concrete technical signals:

- Identifier should be unique, persistent, authoritative, interoperable — a **link**, so notification metadata need not be recopied into every deposition.
- For OBIS/GBIF: **"the BBNJ Identifier could be captured by existing data fields in … Darwin Core, and … incorporated into the occurrence record."** Latimer Core is highlighted for **collection**-level holdings (repositories), not as a substitute for DwC occurrence publishing.
- For INSDC: more development (dedicated UI/metadata field) is expected.
- GGBN permit extensions and Digital Extended Specimen ideas are cited as parallels for propagating legal/administrative tags with samples.
- Caution from ABSCH uptake: machine-readability and adoption matter as much as inventing a string.

A 2026 Frontiers synthesis on CHM capacity adds: EML is routinely **coupled with DwC** in DwC Archives for GBIF/OBIS; CHM success depends on **propagating** identifiers into the repositories where data live; recommend harmonising SBIs with IGSN/SESAR batch approaches; machine-to-machine sync over manual duplicate entry.

3.3 Infrastructure signals

- **OBIS** — GenBank pilot planned; local roadmap already treats batch IDs as a reason to query events/hierarchies efficiently (see also this repo's README).
- **GBIF** — cited as target system; **no** occurrence API parameter or ratified DwC term for BBNJ batch ID. Closest **precedent**: EML/`projectId` at **dataset** grain is searchable in the dataset API, with ongoing debate about **record-level** `projectId` (pipelines #836, rs.gbif.org #115, tdwg/dwc #527) — exactly the same grain problem BBNJ faces.
- **TDWG** — no completed vocabulary enhancement titled for BBNJ batch IDs. DwC-DP generic identifier tables and Latimer Core's typed `Identifier` class are the closest structural hooks.
- **Registered DwC-A extensions** already relevant by analogy: GBIF **Alternative Identifiers** (`Identifier`), **ResourceRelationship**, OBIS **eMoF**, GGBN **Permit**, **dnaDerivedData**.

> **Honest reading of the state of the art**
>
> Literature says "reuse DwC / put it on the occurrence" without naming the term. That is **permission to design a convention**, not evidence that a good convention already exists. Parallel work (GBIF projectId, Latimer typed identifiers, DwC-DP identifier tables) shows the community repeatedly rediscovers: **dataset metadata ≠ record filter.**

4 Design constraints for the query goal

1. **Grain = batch**, often above station/sample → attach to a high-level Event (or Survey), then inherit/denormalise for occurrence query.
2. **Must not destroy** scientific `eventID` / `parentEventID` trees already used in Event-core publishing.
3. **Must be typed** so consumers distinguish CHM IDs from local cruise codes.
4. **Must be indexable** as a first-class occurrence (and ideally event) filter — not only full-text or dataset browse.
5. **Works for DwC-A now** and **DwC-DP next**, ideally via the same projection/lander patterns.

6. **Respect CHM uncertainty**: store opaque strings; validate format only after COP decides.
7. **One string everywhere**: OBIS, GBIF, INSDC, LIMS, CHM notifications.

5 Option analysis

5.1 EML (necessary packaging; insufficient for occurrence query)

OBIS and GBIF already use EML (GBIF Metadata Profile) as the DwC Archive sidecar. Relevant hooks:

- `alternateIdentifier` — list of IDs for the **metadata document / resource** (UUIDs, DOIs, IPT URLs, sometimes cruise codes in practice).
- `project` block — title, personnel, funding, study area; GBIF profile adds a **project identifier** used to group datasets (searchable via dataset API `projectId`).
- Free-text methods / abstract — human-readable only.

What EML is good for

- Associating a **whole published resource** with a batch when the dataset ≈ one cruise/batch (common for dedicated cruise products).
- CHM / DMP packaging: pointing from notification metadata to deposition endpoints.
- Dataset discovery: "show datasets tagged with this project/batch ID."

Why EML is not the solution for the stated goal

- **Wrong grain for many real publications.** Large aggregated datasets (national nodes, multi-cruise compilations, continuous observatories) mix many batches in one EML document. A single `projectId` / `alternateIdentifier` cannot express per-record provenance.
- **Dataset search ≠ occurrence search.** GBIF's `projectId` is a **dataset** facet. Finding datasets is not the same as `occurrenceSearch?bbnjBatchID=…` across the global index — the user goal.
- **No inheritance into occurrences.** Pipelines do not (and should not by default) stamp every occurrence with every EML alternate ID; when they **do** promote metadata (as discussed for projectId), it is an explicit denormalisation decision — which is already an admission that record-level fields are needed.
- **Mixed content.** Cruise codes, DOIs, GBIF UUIDs, and a future BBNJ ID would share undifferentiated identifier slots unless a type/system attribute is conventioned — and even then consumers of occurrence APIs never see it.

> **EML verdict**
>
> **Publish the batch ID in EML too** (typed alternate or project identifier + human note) for FAIR packaging and dataset browse. **Never rely on EML alone** for occurrence/event query in OBIS/GBIF.

5.2 Reusing existing Darwin Core core terms

Rabone et al.'s "existing fields" claim is best read as: **DwC already has identifier-shaped slots; do not invent a parallel data model.** Candidates and verdicts:

Term / pattern	Assessment
`eventID` = BBNJ ID on a parent cruise event	Works mechanically with `parentEventID`; **overloads** scientific hierarchy; collisions with cruise codes; GBIF does not inherit parent fields like OBIS — occurrences often **lose** the ID. Interim only with strict guidance.
`parentEventID` tree only	Hierarchy is necessary; **not** a typed carrier of the legal ID.
`otherCatalogNumbers` / `fieldNumber`	Untyped; polluted semantics; poor discovery.
`dynamicProperties` JSON	Possible interim dump; weakly/ inconsistently indexed; avoid as primary.
`datasetID` / `collectionID`	Wrong grain (publishing unit ≠ collection batch). Same confusion as Movebank/projectId debates.
`associatedSequences` etc.	Wrong direction (links **out** to DSI, does not tag the batch).
eMoF / MoF (`measurementType` = BBNJ ID)	Technically storable; mixes facts with identifiers; awkward as primary key filter; possible **secondary** carrier.

Table 4: Reuse of existing DwC terms for BBNJ batch ID.

Reuse without a **type** discriminator fails the "typed" constraint. Reuse that only lives on a parent event fails the "query occurrences" constraint unless denormalised.

5.3 New Darwin Core term? (probably not — and not BBNJ-named)

A dedicated core term such as `bbnjBatchID` is the instinctive "make it queryable" move. It is also a **large ask** under the TDWG Vocabulary Maintenance Specification (demand, efficacy, stability across the whole DwC community), and a **poor namespace choice** even if a term were justified.

5.3.1 Why a BBNJ-specific term is the wrong ask

- **Treaty-scoped vocabulary ages badly.** DwC is a cross-domain biodiversity standard. Terms named for one instrument (`nagoyaIRCC`, `bbnjBatchID`, …) invite an open-ended series of treaty fields. TDWG has preferred **generic** properties (`recordedByID`, `projectID`, `fieldNumber`) plus controlled values or extensions for jurisdiction-specific documents (GGBN Permit's `permitType=IRCC`).
- **Demand is narrow for a core term.** OBIS/GBIF/INSDC need the **capability**; most DwC publishers never touch ABNJ MGR. Efficacy is easier to show for a reusable pattern (typed identifiers) than for a single-treaty property.
- **Format and governance are still unsettled.** Proposing a core term before CHM string rules exist burns process capital.
- **Index ≠ vocabulary.** OBIS/GBIF can expose an API parameter `bbnjBatchID=` that **reads from** `identifier where type=bbnjBatch` without ratifying a DwC term. Search UX does not require a TDWG local name.

5.3.2 If a term were ever justified — keep it generic

Should typed-identifier rows prove too heavy for mainstream IPT users, a **generic** Event property is the defensible shape, e.g. `collectionBatchID` / `samplingBatchID` for cruise-scale or registry batch tags (SESAR/IGSN, future CHM, similar schemes). BBNJ is then one **value system** in documentation / `identifierType`, not the term name.

`projectID` is already ratified and generic — but its semantics are research/funder projects (Ocean Census, RCN, …), not clearing-house collection batches. Stretching it to BBNJ muddies both use cases.

5.3.3 Why typed identifiers beat a BBNJ-named core term (genuine view)

The product need is an opaque registry string at **batch Event** grain, queryable, shareable with INSDC, and coexisting with a scientific cruise ID. That is multiple **typed** IDs on one event — which DwC-DP `event-identifier` models cleanly. A treaty-named core term (`bbnjBatchID`) still fails when the same event also needs cruise + WIGOS typed IDs, and burns TDWG capital on the wrong namespace.

5.3.4 GBIF Identifier extension: checklist lore vs DwC-DP ingest reality

Registered as `http://rs.gbif.org/terms/1.0/Identifier` (2015) — often called "Alternative Identifiers." Documented properties: `dcterms:identifier` (required), `title`, `subject`, `format`, `datasetID`. Property text historically: "Other known identifier used for the **same taxon**." IPT checklist practice uses it for Taxon-core alternate LSIDs/DOIs/URLs. As **publisher guidance for Event-core DwC-A**, "just use Alternative Identifiers" is therefore a weak recommendation: wrong documented intent, little Event use in the wild, and weak typing in the published schema text.

But GBIF's DwC-DP → DwC-A-shaped ingest already **reuses that rowType** as the landing pad for typed identifier tables. In `gbif/pipelines` (`IdentifierMapping`), fragments such as `eventIdentifiers()` read `event-identifier`, scope by `event_fk`, and emit extension rows of rowType `http://rs.gbif.org/terms/1.0/Identifier`. Parallel fragments pull `survey-identifier`, `occurrence-identifier`, and `material-identifier` into the **same** extension (sometimes routed onto Event, sometimes Occurrence). Allowlisted raw keys include `identifierType` / `identifierTypeIRI`.

So: **mechanically**, Event identifiers can already appear as Identifier extension rows after DwC-DP ingest. **Philosophically / in DwC-A publishing docs**, the 2015 extension is still a checklist artefact. That gap is exactly what a TDWG discussion should resolve — either by **updating** Identifier, or by registering a clean Event Identifier mirror of DwC-DP.

5.4 DwC-A options: update Identifier or new Event Identifier extension

DwC-A has **no** first-class Event equivalent to DwC-DP's `event-identifier` in the **publishing** story. Two credible fixes (both better than a treaty-named core term):

Path	Idea	Trade-off
C1	Update Identifier extension: Event/Occurrence/Material scope; first-class `identifierType` (+ IRI/source); drop "same taxon" wording	Reuses rowType ingest already fans into; repairs a 2015 artefact; name "Alternative Identifiers" stays confusing
C2	Sandbox new Event Identifier extension mirroring DwC-DP (`identifier` + `identifierType` …)	Cleaner mental model; second extension alongside Identifier until pipelines unify
D	Generic Event core term e.g. `collectionBatchID` (not `bbnjBatchID`)	Simple IPT; single-valued; still need cruise as second field; larger VMS ask
E	`bbnjBatchID` core term	Avoid — treaty namespace; invites parallel treaty fields

Table 5: DwC-A paths after Tim Robertson's "generic ID" framing.

Prefer **C1 or C2** in a public TDWG issue framed as **options**, not a finished "New Term - bbnjBatchID" request. Escalate to **D** only if extension+projection fails IPT adoption.

5.4.1 Flat Event/Occurrence field vs `event-identifier` (will the ID get lost?)

The worry is real: **extension tables and DP child tables are the first things dropped** when data are reprocessed — Excel exports, R/`dplyr` pipelines that keep only core columns, GBIF "simple CSV", occurrence-only indexes, LIMS hand-offs, and many IPT republishers who never tick the extension. A column sitting on Event or Occurrence **travels farther by default**.

Carrier	Survives Event keep?	Survives Occ.-only?	Cost
DwC-DP `event-identifier` / DwC-A typed-ID extension	only if extension kept	no (unless projected)	Multiple typed IDs; clean model; easy to drop
Flat field on **Event** only	yes	no — unless inherited	One column; still lost in occurrence-centric tools; weak for BBNJ+cruise+WIGOS together
Flat field on **Occurrence** (denormalised)	n/a	yes	Survives most OBIS/GBIF use; duplicates; wrong grain; hard for multi-ID
Extension **plus** pipeline projection to Event+Occ. search docs	yes (index)	yes (index)	**Best of both:** canonical rows + resilient query surface
Generic core term e.g. `collectionBatchID` on Event (± Occ.)	yes	if also on Occ.	TDWG ask; single-value; still need second field for cruise code

Table 6: Where the batch ID survives common processing paths.

Verdict on the flat-field instinct:

1. **Canonical store** should still be typed identifier rows (DP table / DwC-A extension). BBNJ almost always **coexists** with a scientific cruise ID and sometimes WIGOS/IGSN — one bare column cannot hold that without inventing parallel terms (`bbnjBatchID`, `cruiseID`, `wigosID`, …) or stuffing untyped multi-values into one cell.
2. **Query / hand-off surface** should **not** depend on consumers reading the extension. OBIS/GBIF ingest must **project** `identifier where type=bbnjBatch` onto indexed Event **and** Occurrence documents (API alias `bbnjBatchID=` is fine). That is the same lesson as `projectId` → occurrence search (pipelines #836): metadata that only lives off-core is invisible.
3. **Publisher denormalisation** is optional insurance for flat archives: copy the batch string onto Occurrence (or MaterialSample) **in addition to** the Event-level typed row — documented as denormalised, not as a second source of truth.
4. A **generic** Event core term (`collectionBatchID`) only becomes attractive if, after a real DwC-A extension exists, IPT users still will not attach rows. Do not jump there first: you lose multi-ID and still need Occurrence projection for GBIF-style occurrence search.

So: flat field alone is **more sticky but less expressive**; identifier rows alone are **more correct but fragile**. The product answer is **both layers** — typed rows as source of truth, projected (and optionally denormalised) scalar for survival.

5.4.2 Attachment grain (one parent vs SBI on every sample)

DwC-A is a **star**: each extension row has exactly one `coreid` → one parent core record. DwC-DP `event-identifier` likewise has one `event_fk`. The **SBI string** may still be repeated.

- **Canonical:** one typed-identifier row on the **batch** Event (cruise/leg), `identifierType=bbnjBatch`. Children keep scientific `eventID` / `parentEventID`. Matches Art. 12 batch grain.
- **Allowed denormalisation:** same SBI string on many Occurrence (or station Event) identifier rows — copies for occurrence-centric search/downloads when parent Event extensions are not inherited (GBIF often does not inherit).
- **Not required:** one unique SBI per sample. Many samples share one CHM batch ID.

5.4.3 Should we act on DwC-A before COP freezes the CHM string?

Yes — sandbox C1 and/or C2 soon. Reasons:

- **OBIS (and most marine IPT publishers) still ship DwC-A Event-core.** Waiting for universal DwC-DP adoption leaves a years-long hole exactly where BBNJ pilots need to run. DwC-A → DwC-DP converters exist; reverse packaging remains hard.

- **DwC-DP already defines the structure** ([Event Identifier](https://dwc.tdwg.org/dwc-dp/qrg/#event-identifier)). C2 is a compatibility shim; C1 is documenting/repairing the slot ingest already uses.
- **Motivating types beyond BBNJ** (cruise, WIGOS, IGSN batch, DOI) make the pattern useful even if CHM format stays TBD.
- **Risk if we wait:** overloads of `eventID` / cruise free-text / `projectID` / eMoF calcify.
- **Risk if we ship poorly:** extension never projected → ID "gets lost." Mitigate with `identifierType` vocab + indexing/download commitment in the same pilot.

Process: open a **discussion** issue on `tdwg/dwc` (options A–E; see `reports/bbnj-tdwg-github-issue.md`); sandbox C1/C2 in parallel; do **not** open "New Term - bbnjBatchID" first.

5.4.4 What it should look like

Title: Event / resource identifiers (or "Typed identifiers") — **not** "BBNJ Extension."
Namespace: prefer `http://rs.tdwg.org/dwc/terms/`-aligned or a joint `obis`/`gbif` NS pending TDWG; rowType e.g. `…/EventIdentifier`.
Core: Event (primary); optionally Occurrence / MaterialSample if flat archives need denormalised copies.
Properties (align with DwC-DP `event-identifier.json`):

Property	Req.	Role
`identifier`	yes	The ID string (BBNJ batch, cruise code, WIGOS, …)
`identifierType`	yes	Controlled: `bbnjBatch`, `cruise`, `wigos`, `igsnBatch`, …
`identifierTypeIRI`	no	IRI for the type concept
`identifierTypeSource`	no	Vocabulary / authority URL
`identifierRemarks`	no	Optional human note

Table 7: Proposed DwC-A extension fields (mirror of DwC-DP event-identifier).

BBNJ specificity lives **only** in the type vocabulary value, not the extension name. IRCC stays on GGBN Permit (`permitType` + `permitURI`) on MaterialSample — different legal object.

5.4.5 How it would be published

1. Event-core DwC-A (OBIS normal path).
2. One extension row per ID on the **batch/cruise** event (`identifier` + `identifierType=bbnjBatch`).
3. Scientific `eventID` / `parentEventID` hierarchy unchanged; station/sample children keep their own `eventID`s.
4. Ingest: event index stores typed IDs; recurse hierarchy; join occurrences on `(dataset_id, eventID)`; **always** project `identifier` where type=`bbnjBatch` onto occurrence search docs / API alias (do not leave the value extension-only).
5. Optional publisher denormalisation: same string on Occurrence/MaterialSample for flat CSV consumers — marked as copy of the Event-level typed ID.
6. Molecular: same batch string on BioSample attribute; `materialSampleID` = BioSample accession; `associatedSequences` as today.
7. EML may repeat the ID for packaging — not the query path.

5.4.6 Community process

1. **Discussion issue** on `tdwg/dwc` (options A–E; draft in `reports/bbnj-tdwg-github-issue.md`) — not a finished `bbnjBatchID` term request.
2. Choose C1 vs C2 with GBIF/TDWG; draft extension/vocab XML; validate against `extension.xsd`.
3. PR to `gbif/rs.gbif.org` **sandbox**; IPT test; synthetic pilot (batch Event → children → BioSample).
4. Pipelines: project typed ID to Event+Occurrence indexes; download layouts as above.

5. Promote to production; publisher docs (OBIS/GBIF data blog / IPT notes).
6. Same type string in BioSamples pilot; revisit generic core term **only** if extension+projection fails adoption.

5.4.7 Fallbacks

- **DwC-DP-only:** if OBIS moves Event publishing to DwC-DP quickly, skip the DwC-A extension and use `event-identifier` natively.
- **Generic core term `collectionBatchID`:** simpler single column, but weak for multiple concurrent typed IDs on one event — inferior to the extension/DwC-DP table for BBNJ+cruise+WIGOS.
- **Provisional OBIS term URI on Event core** (custom IPT mapping): fastest pilot hack; acceptable for a prototype, not a multi-network standard.

5.4.8 ResourceRelationship / eMoF / dnaDerivedData / Identifier-as-is

Not primary homes without repair. Using today's Alternative Identifiers **unchanged** (checklist wording, no documented Event `identifierType`) is insufficient — even though DwC-DP ingest already fans into that rowType.

> **Verdict**
>
> **Canonical:** DwC-DP `event-identifier` with `identifierType=bbnjBatch`. **DwC-A:** update Identifier (C1) **or** sandbox Event Identifier mirror (C2). **Resilience:** project onto Event+Occurrence indexes; downloads: dedicated identifier table + optional Occurrence column. No treaty-named core term; GGBN Permit for IRCC.

6 How users see the ID in downloads

Publishing and indexing are not enough if a Darwin Core or Parquet download still hides the batch ID. Treat download layout as part of the product.

6.1 What not to do

- **Extension-only, no projection:** Identifier / Event Identifier file present in the zip, but simple CSV / occurrence.txt / R one-liners never see it — same failure mode as unindexed `projectId`.
- **BBNJ-only sidecar that replaces the generic identifier table:** works for a pilot, but breaks the multi-type story (cruise + BBNJ + WIGOS) and diverges from DwC-DP.
- **Rely on Event inheritance alone in GBIF-shaped occurrence downloads:** parent Event extensions often do not appear on child occurrences.

6.2 Recommended download shapes

Product	Presentation
DwC-A zip (Event-core)	Core `event.txt` + **generic** identifier extension file (`identifier.txt` / Event Identifier) with `coreid` → batch Event. Occurrences join via `eventID` / `parentEventID`. Optionally add a **denormalised** column on `occurrence.txt` (e.g. interpreted `bbnjBatchID` or a documented dynamic field) for simple consumers.
DwC-A zip (Occurrence-core / flat)	Either denormalise SBI onto every occurrence row **or** ship identifier extension rows with `coreid` = occurrence `id` (repeated SBI string). Prefer also linking back to a batch `eventID` when Event hierarchy exists.
DwC-DP zip	`event-identifier` table (and `survey-identifier` if Survey-modelled). Do **not** invent a BBNJ-only table. Same string may appear on child events only if denormalised.
OBIS multi-table Parquet	`event` + extension/identifier parquet; DuckDB walk from filtered occurrences → parent events → identifier rows where `identifierType=bbnjBatch`. Optional: materialise `bbnjBatchID` on occurrence parquet for one-shot filters (projection, not source of truth).
GBIF simple CSV / species list	Must include a projected column if this format stays popular — otherwise BBNJ is invisible. Long-term: push users to DwC-A/DP or SQL downloads that retain the identifier table.
GBIF SQL / DwC-DP download (future)	Expose identifier table or a clear `bbnjBatchID` / typed-identifier view; document join to `gbifID` / `eventID`.

Table 8: How to present BBNJ batch IDs in user downloads.

6.3 Practical default for OBIS pilots

1. **Source of truth:** one typed-identifier row on the batch Event.
2. **Search:** project onto Event + Occurrence documents (API alias `bbnjBatchID=` fine).
3. **Full download:** always include the identifier table/file; do **not** make "every occurrence row carries SBI" mandatory in the relational export.
4. **Simple / occurrence-only download: yes**, add the SBI to every included occurrence (denormalised column or repeated extension row) so spreadsheet users are not forced to join.
5. **Do not** ship a separate `bbnj.txt` that invents treaty-only columns unless CHM workflow fields are explicitly mirrored — and even then prefer CHM URLs in EML / optional sparse fields over a parallel standard.

> **Download bottom line**
>
> Relational download = dedicated identifier file (generic). Simple occurrence download = denormalised SBI on each row. Both read from the same Event-level typed source. Never BBNJ-only as the **only** carrier.

6.4 Hypothetical: if there were a BBNJ-specific extension

This subsection is **not** the recommended path (prefer generic typed identifiers). It answers: **what fields would a treaty-branded extension reasonably contain?** Use it as a requirements checklist — many fields belong in the CHM or EML, not on every DwC row.

Attach primarily to **Event** (batch/cruise); optional denormalised copy on Occurrence/MaterialSample for flat archives. 1 :many if one event participates in several notifications (unusual).

6.4.1 Core identity (minimum viable)

Field	Req.	Notes
`bbnjBatchID`	yes	Opaque CHM-issued string. Format TBD — treat as string, no regex until COP decides.
`bbnjBatchIDURI`	no*	Resolvable CHM record URL. *Required once CHM resolution exists.
`bbnjIdentifierScheme`	no	Fixed or vocab: `BBNJ-CHM` — future-proofs if multiple issuers/ versions.

Table 9: Hypothetical BBNJ extension — identity fields.

6.4.2 Linkage to CHM workflow (optional, sparse)

Field	Req.	Notes
`bbnjNotificationStage`	no	Vocab: `preCollection` / `postCollection` / `utilization` — usually **wrong** on biodiversity deposits (those are CHM records). Include only if the DwC row is explicitly a notification mirror.
`bbnjPreCollectionURI`	no	Link to pre-collection notification (art. 12(2)–(3)).
`bbnjPostCollectionURI`	no	Link to post-collection update (art. 12(5)).
`bbnjUtilizationURI`	no	Link to utilization notification (art. 12(8)), if any.
`bbnjDMPURI`	no	Data management plan URI from notification — often better in EML `project` / additionalMetadata.

Table 10: Hypothetical BBNJ extension — CHM workflow fields.

6.4.3 Provenance / deposit hints (optional)

Field	Req.	Notes
`bbnjMGRRepository`	no	Where physical MGR are/will be held (art. 12(5)) — name or URI.
`bbnjDSIDatabase`	no	Where DSI is/will be recorded (e.g. INSDC) — prefer also `associatedSequences` / BioSample.
`bbnjAppliesToRecord`	no	Vocab: `mgr` / `dsi` / `biodiversity` / `all` — for mixed cruises where not every occurrence is MGR.
`bbnjGeographicScope`	no	Free text or URI; coordinates stay in DwC Location terms.
`bbnjRemarks`	no	Human notes; never the only carrier of the ID.

Table 11: Hypothetical BBNJ extension — deposit and scope fields.

What not to stuff into a BBNJ extension: full art. 12(2) narrative (objectives, vessel tonnage, participation opportunities) — that is CHM/EML content. Scientific cruise hierarchy — `eventID` / `parentEventID`. Nagoya IRCC — GGBN Permit. Sequence accessions — `associatedSequences` / DNA-derived data.

Lean hypothetical profile for pilots: require only `bbnjBatchID`; add `bbnjBatchIDURI` when resolvable; everything else optional or left in CHM.

6.5 Draft / sandbox extensions without a definitive legal basis

CHM string format, resolution API, and exact batch grain are still COP follow-ons. That does **not** block technical prototyping — GBIF's schema registry is built for iterative, pre-production definitions.

6.5.1 What "sandbox" means here

- Host: `https://rs.gbif.org/sandbox/` (`extension/`, `vocabulary/`, also `experimental/`).
- Listed on sandbox `extensions.html` / `extensions.json` — **not** the production registry IPT instances use by default.
- Purpose (IPT manual): communities **prototyping** new cores/extensions, mapping data, refining properties before freezing.

- Production promotion: only when the property set is stable; at GBIF Development team discretion; sandbox traces removed on move to production (`versioning.md`).

Sandbox ≠ TDWG ratification. Sandbox ≠ CHM legal validity. It is an engineering quarantine.

6.5.2 Recommended process while law/format is unsettled

1. **Write a short intent note** — problem (query by batch ID; link to INSDC), non-goals (not replacing CHM; not Nagoya IRCC), and explicit "format TBD / synthetic IDs."
2. **Choose scope** — prefer **generic** typed-identifier extension; if exploring a BBNJ-branded draft, mark description as **VOLATILE / experimental** (precedent: UK NBN sandbox-style extensions flagged volatile).
3. **Author XML** — comply with `http://rs.gbif.org/schema/extension.xsd`; date-stamp filename (`…_YYYY-MM-DD.xml`); `dc:issued`; namespace under a non-production URI if needed (e.g. `http://rs.iobis.org/obis/terms/` or sandbox terms path).
4. **Vocabularies** — separate thesaurus XML for controlled values; keep `bbnjBatchID` format **unconstrained** (plain string) until COP publishes a pattern.
5. **PR to `gbif/rs.gbif.org`** — land under `sandbox/extension/` (and `sandbox/vocabulary/`); do **not** ask for production.
6. **Configure a test IPT** — point at sandbox registry / install the draft extension; map one synthetic dataset (fake batch IDs are fine).
7. **Iterate in sandbox** — adding/removing properties requires **new dated versions** (immutable extension policy); descriptions/examples can change without a new version.
8. **Align pilots** — GenBank/BioSamples use the same synthetic strings; document that IDs are non-authoritative.
9. **Stay out of production** until: (a) CHM can issue or resolve real IDs, or (b) the community agrees a **generic** typed-identifier extension is stable enough without waiting on BBNJ format.
10. **Optional parallel:** DwC-DP `event-identifier` examples in a data package — no IPT extension registry needed for pure DP trials.

6.5.3 Rules of engagement without legal basis

- **Do** prototype indexing, joins, and `resolve(batch_id)` with synthetic IDs.
- **Do** keep properties optional except the ID string; avoid validators that assume a CHM regex.
- **Do not** present sandbox terms as CHM-compliant or Party obligations.
- **Do not** freeze production vocabulary on guessed syntax.
- **Do not** conflate sandbox publication with TDWG DwC standard status.
- When COP decides format: issue a **new extension version** (or migrate values into the generic typed-identifier extension) and remap; IPT upgrade path is designed for that.

> **Sandbox takeaway**
>
> No definitive legal basis is a reason to stay in **sandbox / experimental** and use **synthetic IDs** — not a reason to do nothing. Prefer sandboxing a **generic** typed-identifier extension; a BBNJ-named draft is only for exploring field lists, and should stay volatile until CHM exists.

6.6 DwC-DP identifier tables (native for DP)

6.6.1 What `event-identifier` is

`event-identifier` is a DwC Data Package **child table**: one row = one **typed** external (or alternate) identifier **for** a `dwc:Event`. It is not a replacement for `event.eventID`. The Event table still has its structural `eventID`; this table hangs additional IDs off that key.

Where it comes from: DwC-DP (TDWG / GBIF data-package model under `tdwg/rs.tdwg.org` and `gbif/dwc-dp`). The table's formal type is `adms:Identifier` — the same pattern is repeated for `survey-identifier`, `occurrence-identifier`, `organism-identifier`, `material-identifier`, `media-identifier`, `agent-identifier`. So "typed identifiers beside the core record" is a DwC-DP design motif, not a BBNJ special case.

Columns that matter:

Field	Role
`eventID` (required)	Foreign key **to** the Event row this identifier is **for** (`predicate: for`). Not unique — many identifier rows may share one `eventID`.
`identifier` (required)	The ID string itself (`skos:notation` in the RDF mapping).
`identifierType`	Human/controlled category of the scheme (e.g. cruise, DOI, future `bbnjBatch`).
`identifierTypeIRI` / `identifierTypeSource`	Machine type + vocabulary/authority for that type.
`identifierLanguage`	Optional language of the identifier presentation.

Table 12: DwC-DP `event-identifier` fields (from the table schema).

Intended use: keep `event.eventID` as the dataset join key for `parentEventID` / occurrences / extensions; put **registry and campaign labels** (CHM batch ID, scientific cruise code, WIGOS, IGSN batch, …) as separate typed rows. Pipelines can project `identifier` where `identifierType = bbnjBatch` onto search documents without overloading the structural key.

Multiple identifiers? Yes. `eventID` is required but **not** unique on this table; the foreign key is many :1 (many identifier rows → one Event). One cruise Event can carry BBNJ + cruise code + DOI as three rows.

6.6.2 What ADMS is

ADMS (Asset Description Metadata Schema) is a W3C vocabulary for describing semantic assets (schemas, code lists, …) in catalogues. DwC-DP reuses only its **`Identifier` class** — based on the UN/CEFACT Identifier pattern: content string + optional scheme / scheme version / issuing agency. In RDF, the string is `skos:notation`; properties of `adms:Identifier` describe the **identifier**, not the Event. That is why DwC-DP separates "the Event" from "IDs that point at the Event," and why `identifierType*` fields live on the identifier row.

Publish the batch ID as a typed row on the cruise/batch-level event (or use `survey-identifier` if the batch is modelled as a Survey). This is the cleanest structural fit and **avoids** a treaty-specific core term entirely.

6.7 Latimer Core (complementary)

Latimer Core's typed `Identifier` class at **collection / organisational unit** grain matches repository holdings and transfers (Rabone et al.). It does **not** replace DwC occurrence/event publishing into OBIS/GBIF. Use both: LtC for collection registries; DwC for occurrence query.

6.8 Suggested architecture

CHM issues BBNJ batch ID
↓
EML (optional): alternateIdentifier / project note
and
Batch/cruise Event + typed identifier
(DwC-A Identifier ext. / DwC-DP `event-identifier`;
optional GGBN Permit for IRCC on materials)
↓ `parentEventID` / `event_fk`
Station / sample events → occurrences / materials / sequences
↓ ingest projection
Search document: denormalised field or alias
(API may call it `bbnjBatchID` even if DwC does not)

Figure 1: EML packaging; typed ID on batch Event; search alias may be BBNJ-named without a DwC term.

7 Decision matrix (query goal)

Option	Typed	Occ. filter	Effort	Role
EML only	weak	no	low	Packaging / dataset browse
EML + promote to all occ.	weak	yes*	med	*Only if 1 batch/dataset; brittle
`eventID` overload	no	partial	low	Interim anti-pattern
Untyped `otherCatalogNumbers`	no	poor	low	Avoid
eMoF as ID	via vocab	awkward	med	Secondary only
Identifier extension / `event-identifier` + index	yes	yes	med	**Best default — no new DwC term**
Thin multi-field extension (GGBN-shaped)	yes	yes	med	If CHM needs status/URL fields
Generic `collectionBatchID` term	yes	yes	high	Fallback only
`bbnjBatchID` DwC core term	yes	yes	very high	Avoid — treaty-specific
Latimer Core only	yes	no	med	Collections, not OBIS search

Table 13: How well each option supports occurrence query in OBIS/GBIF.

8 Making it queryable in OBIS and GBIF

Publishing without indexing is theatre. Even perfect DwC still fails if the global search document lacks a dedicated field.

8.1 OBIS

Publish: Event-core (or DwC-DP) with typed batch ID on the highest relevant event; document the field/type for publishers; also record in EML for the resource.
Ingest: Prefer the **event-first** path enabled by the separate-tables prototype (event index + `event.parquet`): index the typed batch ID on events; resolve child events via `parentEventID`; join occurrences / DNA / MoF on `(dataset_id, eventID)`. Optionally still copy the batch ID onto occurrence search documents for one-shot occurrence filters — that denormalisation is a UX optimisation, not a substitute for an event store.
API/UI: Explicit parameter and facet (`bbnjBatchID` / agreed name) on **both** event and occurrence search; do not rely on full-text.
Pilot: Align with GenBank so one string retrieves OBIS events/occurrences and INSDC sequences.

8.1.1 Why the event index + `event.parquet` prototype helps

The README prototyping model (Event index; per-dataset `event.parquet` + occurrence/extension tables; DuckDB joins on `dataset_id` + DwC IDs; field inheritance for filtering) matches the BBNJ grain unusually well:

1. Batch IDs attach at **cruise/batch** events, not at every station/sample. An event index can filter those parents directly without stuffing the ID into every occurrence document up front.
2. `query_subset.py`-style recursion already expands `eventID` / `parentEventID` keys, then exports related occurrences and extensions — the same pattern yields "all samples/occurrences under this batch."
3. Downloads stay non-duplicative: return the event subset + occurrence subset + DNA/MoF tables; clients join. That is exactly what you want for a batch-scoped MGR package.

Critical distinction: use the batch ID as a **filter field on the event** (typed term / identifier row / inherited interpreted field). Use `eventID` only as the **join key** to child events and occurrences. Do **not** set `eventID = BBNJ batch ID` unless you accept the scientific-hierarchy overload discussed above. With inheritance on interpreted fields, a child sample event can be filterable by the parent's batch ID while keeping its own station-level `eventID`.

So: yes — event index + separate event Parquet makes "get everything for this batch" **easier and cleaner** than occurrence-only indexing. It does not remove the need for a typed batch-ID field; it removes the need to pretend the batch ID **is** the `eventID`.

8.2 GBIF

Publish: Same convention via IPT (core term and/or Identifier/BBNJ extension; EML project/ alternate for packaging).
Interpret: Map typed identifier into the occurrence index as a dedicated interpretable field (learn from the `projectId` promotion debate — do it **explicitly** for BBNJ).
API: Predicate download / occurrence search parameter; SQL downloads as a column.
Caveat: Weaker event-inheritance than OBIS strengthens the case for denormalisation at publish **or** interpret time.

8.3 Cross-system consistency

Agree one opaque string (then CHM format); prefer resolvable IRIs if offered; publish `identifierType` / term IRI so pipelines recognise BBNJ among cruise codes; keep scientific cruise IDs as **separate** typed identifiers. The same string must be what INSDC/BioSamples store — see next section.

9 INSDC, EBI BioSamples, and the linking problem

Humphries et al. state the architectural intent plainly: a GBIF (or OBIS) record and an INSDC record that share the same BBNJ Identifier **are linked by that ID**, analogous to how BioProject groups related submissions. Building a system that **uses** that link is a separate engineering problem from choosing a DwC term.

9.1 What INSDC has signalled (and what it has not)

From the consolidated CHM technical study (2026) and Rabone et al. (2025):

- INSDC is preparing to **accept** the batch identifier using **free-text metadata fields where submitters currently enter cruise identifiers**.
- INSDC would **not enforce** the requirement; compliance rests with the submitter (and ultimately Parties).

- More development may be needed for a **dedicated UI / explicit field** on DSI pages — free-text cruise slots are an interim, not necessarily the end state.
- **OBIS is planning a pilot with GenBank** to demonstrate technical linking of scientific data and DOIs across systems via the batch identifier.
- OBIS's public BBNJ note (Nov 2025) further argues SBIs should be globally unique and persistent, and floats **DOI-based** SBIs as one way to power citation and provenance — a design choice that would also affect INSDC and DwC string patterns if COP adopts it.

Not decided: attribute local name on BioSamples; whether it becomes a checklist-mandatory field (like recent spatiotemporal minimum standards) or remains optional free text; GenBank flat-file `/source` qualifier vs BioSample-only; NCBI vs ENA vs DDBJ UI parity; search indexing of the attribute in ENA Portal vs BioSamples `attr:` filters only.

9.2 What EBI / MIxS have (and have not) done — GitHub check

EMBL-EBI sits in this story as the European INSDC node (ENA) and as host of BioSamples and the OLS mirror of GSC MIxS — **not** as an independent BBNJ CHM implementer. A deliberate search of public GitHub and GSC issues (Sep 2026) finds **no** EBI/`enasequence`/`ebi` repository that implements, prototypes, or documents a BBNJ batch-identifier field. The only concrete INSDC signal remains the CHM study's line about **cruise free-text** reuse; the OBIS–**GenBank** pilot named in that study is NCBI-facing, not an ENA code drop.

What **does** exist in the MIxS / GSC track is adjacent vocabulary work — and one dangerous name collision:

Term / issue	Status	Relevance to BBNJ BSBI
`batch_ids` (MIXS:0001382); mixs#1160	Accepted; in current schema; Ancient / MInAS	**Wrong semantics.** Lab / processing umbrella IDs (extract, library, seq run, sediment core) for **batch-effect** analysis — not a CHM legal collection batch. Do **not** overload this for BBNJ.
`permit_id` / `permit_authority` / `permit_date` / `permit_scope` (MIXS:0001364+); mixs#1141–1144	Accepted; Ancient / MInAS	Generic ethical/legal approval IDs. Could **carry** a BBNJ string as free text, but typed discovery and cruise/batch semantics are wrong.
`ethical_id`; mixs#573	Closed; MInAS	Predecessor framing for permit-style IDs.
`abs_certificate` / ABSCH UID; mixs#20 (2018–19)	Approved at GSC CIG calls; **absent** from current `mixs.yaml` / OLS	Nagoya IRCC / ABSCH unique identifier — **not** BBNJ. Shows GSC once agreed an ABS term and then **failed to ship it** into the live vocabulary. Cautionary precedent for hoping MIxS will spontaneously mint a BBNJ term.

Table 14: GSC/MIxS terms adjacent to BBNJ — none are a BBNJ standardized batch identifier.

Bottom line on a MIxS term: there is **no** evidence that EBI or GSC intend to create a dedicated BBNJ / BSBI MIxS slot. The live `batch_ids` term would be a **harmful** reuse target. The closest shipped pattern is `permit_*` (wrong grain/type), and the closest historical ABS work (`abs_certificate`) never landed in the published schema. For ENA/BioSamples, the practical path remains: (1) short-term cruise free-text as INSDC announced; (2) push for a **named, indexed BioSample attribute** (checklist or recognised optional key) aligned with the DwC `identifierType` string — via INSDC/GSC process, not by hijacking MIXS:0001382.

Non-EBI GitHub that **is** BBNJ-related: iobis/bbnj-chm (CHM prototype project management — MGR/EIA modules, APIs, metadata fields; not sequence checklists) and older iobis/bbnj ABNJ dataset lists. Treat those as CHM/OBIS planning, not EBI deliverables.

> **Why INSDC's choice cascades into DwC**

If INSDC standardises a **named, indexed** BioSample attribute (ideal), DwC should use the **same local name / type IRI** so joins are trivial. If INSDC stays on undifferentiated cruise free-text, DwC **still** needs a typed field — and the linking system must parse/normalise noisy cruise strings, which is fragile. Do not wait passively: propose a joint attribute name to INSDC/EBI while designing DwC, and treat the GenBank–OBIS pilot as the negotiation venue.

9.3 Where the ID should live in the sequence stack

INSDC is a stack, not a single table. Placement matters for query and join:

Object	Grain	Role for linking
BioSample (`SAMN`/`SAMEA`/…)	Physical / material sample	**Best primary home.** One sample → many runs/sequences. BioSamples API supports `filter=attr:field:value`. Checklist attributes index better than ad-hoc free text.
BioProject / Study	Study / programme	Useful when one project = one batch; fails for multi-batch projects. Analogous to EML `projectId` pitfalls.
Run / Experiment (SRA/ENA)	Sequencing assay	Too fine; inherit from sample.
Nucleotide accession (GenBank flat file)	Sequence record	Human-visible; `/source` notes or structured qualifier if INSDC adds one. Secondary copy of sample-level ID.
Cruise free-text (current practice)	Varies	Today's slot INSDC cites; untyped; collisions with scientific cruise codes — same anti-pattern as overloading `eventID`.

Table 15: Candidate homes for the BBNJ batch ID in INSDC/EBI.

State of the art for marine molecular submissions already puts rich campaign/station metadata on **BioSamples** (e.g. Tara Oceans attributes). That is the natural place for a typed `bbnj_batch_id` (name TBD) — ideally via an ENA checklist update or a recognised optional attribute, not only submitter-defined keys that Portal search may not return.

9.4 Existing DwC ↔ sequence bridges (reuse these)

Do not invent a parallel molecular model. Current best practice already defines the **edges** of the link graph:

- `materialSampleID` — **use the BioSample accession** when one exists (GBIF/OBIS DNA guidance).
- `associatedSequences` — URLs/accessions for BioProject, nucleotide, or SRA/ENA records (DwC core; indexed enough for discovery when populated).
- **DNA-derived data extension** — MIxS-aligned method metadata; links via `occurrenceID` / `eventID`; OBIS indexes more of this path than GBIF's portal (GBIF often leaves extension content to source-archive download).
- Future DwC-DP material / sequence tables — richer many-to-many without star-schema duplication.

The **missing node** is the shared batch ID on both sides. The **missing service** is a resolver that walks batch ID → {OBIS/GBIF occurrences, BioSamples, accessions, CHM notification}.

9.5 Architecture for a linking system

Target capability: given a BBNJ batch ID (or a BioSample / occurrence / accession), return the connected set across CHM, OBIS/GBIF, and INSDC/EBI.

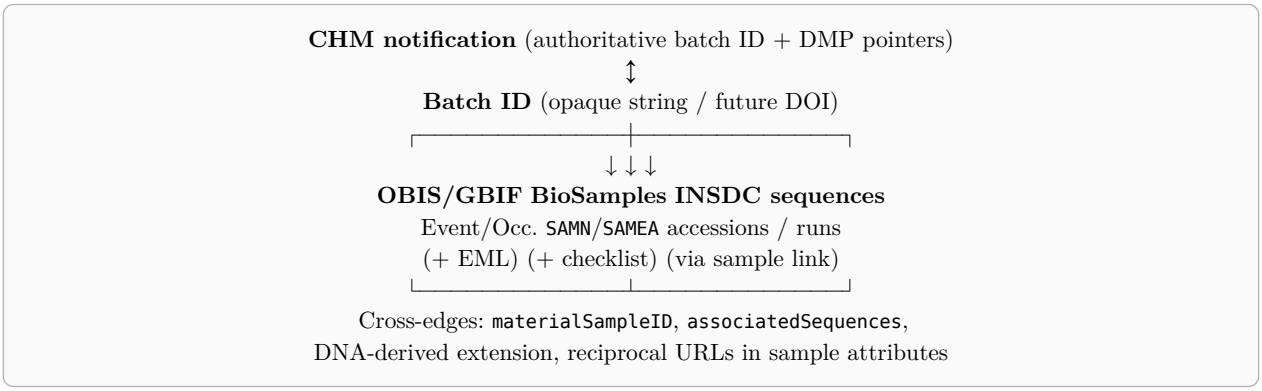


Figure 2: Link graph centred on the batch identifier.

9.5.1 Link modes (align with CHM study)

The CHM study's interoperability modes map cleanly onto what OBIS/GBIF should build:

1. **Link mode (baseline):** Stable URLs and typed IDs on record pages — CHM ↔ dataset/occurrence ↔ BioSample ↔ accession. Lowest burden; sufficient for human navigation and simple scripts.
2. **Harvest mode:** Periodically pull BioSamples/ENA metadata (and OBIS/GBIF occurrence docs) into a **batch-ID index** so one query returns cross-system hits without live federation. Fits DuckDB/ Parquet lakes and Elasticsearch sidecar indices.
3. **API federation:** Authenticated joins for CHM aggregate reporting and Party dashboards — higher value, higher governance cost.

For an OBIS-led product, **harvest + link** is the pragmatic MVP; federation can wait for CHM APIs.

9.5.2 Join keys by grain

Grain	Join
Batch	Identical BBNJ string in DwC typed field **and** BioSample attribute (and CHM). This is the **only** key that works when sequences and occurrences were published by different teams.
Sample	`materialSampleID` = BioSample accession; also store batch ID on both sides for redundancy.
Sequence	`associatedSequences` / DNA extension → accession; accession → BioSample → batch ID.
Dataset	EML project/alternateIdentifier; BioProject ID — weak alone, useful as a hint.
DOI	If SBI is DOI-based (OBIS proposal), Content Negotiation / DOI API becomes a fourth resolver.

Table 16: Primary join keys between systems.

9.5.3 What the linker must normalise

Until INSDC and DwC converge on one attribute:

- Accept multiple source fields (typed DwC term, Identifier-extension type, BioSample `attr:*`, cruise free-text with heuristics, EML project ID).
- Emit a **canonical** batch ID record: `{id, id_type, sources[], occurrence_ids[], biosample_accessions[], sequence_accessions[], chm_url?}`.
- Flag low-confidence links (cruise-string fuzzy match) separately from high-confidence typed matches.
- Support **multiple batch IDs** per sample/occurrence if COP allows composite projects.

9.5.4 Implications for DwC publishing (molecular datasets)

For ABNJ molecular work published to OBIS/GBIF:

1. Put typed `bbnjBatchID` on the batch/cruise Event; denormalise to occurrences.
2. Set `materialSampleID` to the BioSample accession whenever sequences exist.
3. Fill `associatedSequences` with resolvable BioProject/SRA/nucleotide URLs.
4. Use DNA-derived data extension for MIxS method fields (already aligned with ENA checklists).
5. Mirror the batch ID into BioSample attributes **with the agreed attribute name** at submission time — do not rely on later text-mining.
6. Keep EML packaging in sync for the DwC Archive.

That is the publishing contract the linking system can rely on.

9.5.5 Pilot scope (OBIS–GenBank)

A minimal successful pilot:

- One synthetic (or consented real) batch ID.
- Matching BioSample(s) in GenBank/ENA with the ID in a documented attribute.
- Matching Event-core DwC-A in OBIS with typed ID + `materialSampleID` + `associatedSequences`.
- A small service or notebook: `resolve(batch_id) → {obis_occurrences, biosamples, accessions}`.
- Documentation of field names on **both** sides — feed that into TDWG / INSDC checklist discussions.

Failure mode to avoid: pilot only demonstrates DOI minting or only DwC publishing, without a round-trip query from batch ID through both infrastructures.

10 Recommended path (actionable)

1. **Open TDWG discussion** on `tdwg/dwc` (options; see `reports/bbnj-tdwg-github-issue.md`) — not "New Term - bbnjBatchID."
2. **DwC-DP:** publish via `event-identifier` (`identifierType=bbnjBatch`).
3. **DwC-A:** sandbox **C1** (update Identifier) and/or **C2** (Event Identifier mirror); do **not** recommend Identifier-as-is; do **not** wait for COP format.
4. **Index:** project typed batch ID onto event **and** occurrence search docs; join hierarchy on scientific `eventID`.
5. **Downloads:** identifier table in full/relational exports; denormalised SBI on simple occurrence exports.
6. **INSDC/BioSamples:** same type string in the GenBank–OBIS pilot — not cruise free-text long-term.
7. **GGBN Permit** for IRCC when Nagoya applies; MIxS carries the string beside checklists but is not the home.
8. **Pilot** `resolve(batch_id)` before wide rollout.
9. **Fallback only:** generic `collectionBatchID` core term if extension+projection still fails IPT adoption.

11 Open questions

1. Final CHM string format — opaque token vs DOI — and resolution API?
2. Grain: cruise vs leg vs deployment vs project?
3. Multiple batch IDs on one occurrence/sample?
4. Tag non-MGR biodiversity from the same cruise?
5. Who validates published IDs against the CHM?
6. IGSN/SESAR relationship — alias, namespace, or independent?
7. Will INSDC promote a **checklist** attribute or stay on cruise free-text?
8. NCBI / ENA / DDBJ parity for UI and search indexing?

9. Will GSC ever mint a BBNJ-specific MIxS slot — or only reuse `permit_*` / free text? (Do not use `batch_ids`.)
10. Will GBIF index DNA-extension / associatedSequences richly enough for reverse links?
11. Who operates the harvest-mode link index?
12. C1 (update Identifier) vs C2 (new Event Identifier) — which does TDWG/GBIF prefer?
13. Download policy: always denormalise SBI onto occurrence simple CSV, or only when Event hierarchy is absent?

12 Conclusions

1. The BBNJ batch ID is a batch-level provenance key — not a scientific DwC ID.
2. **Best solution:** typed identifier rows (DwC-DP; DwC-A via C1 or C2) **plus** pipeline projection so the ID does not vanish in occurrence-centric processing.
3. Act on DwC-A **now** (sandbox); do not wait for universal DwC-DP or final CHM syntax; open a **discussion** issue first.
4. Identifier-as-is is unsuitable as publisher guidance; updating it is a valid path because DwC-DP ingest already fans `event-identifier` into that rowType.
5. No BBNJ-specific DwC core term; flat generic core field only as fallback.
6. Downloads: dedicated identifier file for relational exports; denormalised SBI on simple occurrence downloads.
7. Do not overload `eventID` / `projectID`; attach canonically to the batch Event; denormalise to samples only for search/UX.
8. Spend effort on TDWG consensus, indexing/projection/downloads, and the GenBank–OBIS pilot.

13 References

1. BBNJ Agreement (A/CONF.232/2023/4), arts. 12, 14, 16.
2. Consolidated draft study on technical aspects of CHM operationalisation (2026) — OBIS–GenBank pilot; INSDC free-text approach.
3. PrepCom note on CHM issues (2025).
4. UK guidance — BBNJ collection of MGR in ABNJ.
5. Humphries et al., "Data Management and the 'BBNJ Standardized Batch Identifier'…" in **Decoding Marine Genetic Resource Governance** (Springer, 2025) — doi:10.1007/978-3-031-72100-7_12.
6. Rabone et al., "BBNJ Agreement: Considerations for Scientists and Commercial End Users…" (Springer, 2025) — doi:10.1007/978-3-031-72100-7_14 — existing DwC fields; Latimer Core; INSDC gaps.
7. Related Springer chapters on notification systems and monitoring/transparency (same volume).
8. Frontiers (2026) — data-centric capacity for the BBNJ CHM — EML+DwC coupling; identifier propagation; IGSN/SESAR harmonisation.
9. Harvard BBNJ MGR Mechanism explainer.
10. OBIS — digital foundation for BBNJ / SBI & DOI discussion (Nov 2025).
11. OBIS manual — DNA-derived data GBIF — publishing DNA-derived data — `materialSampleID`, `associatedSequences`, DNA extension.
12. INSDC spatiotemporal minimum standards — precedent for BioSample checklist mandates.
13. EBI BioSamples attribute filters — `attr:field:value` search.
14. ENA sample checklists / Portal API — indexed checklist attributes vs submitter-only free text.
15. GSC MIxS: `batch_ids` #1160 (MIXS:0001382 — lab batches, **not** BBNJ); `permit_id` #1141–1144; `abs_certificate` #20 (approved 2019, never shipped).
16. iobis/bbnj-chm — BBNJ CHM prototype project management (not EBI).
17. OBIS manual — EML Event format.
18. GBIF Metadata Profile (EML) — `alternateIdentifier`, project block / project ID.
19. DwC `projectID` / `projectTitle` — tdwg/dwc#532 (completed); Humboldt survey guides — keep project IDs and `fieldNumber` out of `eventID`.
20. gbif/dwc-dp#159 / #150 — structural vs semantic keys; #112 site identifiers.
21. gbif/pipelines#836 / rs.gbif.org#115 — record-level `projectId` indexing.
22. GGBN Permit Vocabulary / Permit extension — `permitType` (incl. IRCC), `permitStatus`, `permitURI`; GGBN wiki.
23. ABSCH / Nagoya IRCC guides — certificate UID; publishing IRCC uptake critiques in Rabone et al. 2019, Rohden et al. 2020, Scholz et al. 2022, Humphries et al. 2025.
24. Humphries et al. 2021 — traceability approaches for MGR under BBNJ — ABSCH vs GGBN permit standards.
25. GBIF registered extensions — Identifier, ResourceRelationship, eMoF, GGBN Permit, dnaDerivedData.
26. Darwin Core terms TDWG Vocabulary Maintenance Specification.
27. IPT — add a new core/extension rs.gbif.org versioning — sandbox → production; immutable versions.
28. GBIF schema sandbox — draft extensions/vocabularies.
29. Latimer Core — typed collection-level identifiers.

Briefing based on public treaty, PrepCom, scholarly, and standards sources (Sep–Oct 2026). Revised 19 Oct 2026: TDWG options framing; DwC-DP `IdentifierMapping` fan-out; download presentation. CHM modalities and identifier format remain subject to COP decisions. Typst report.